An Intelligent LEGO Tutoring System for Children with Autism Spectrum Disorder

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ABSTRACT
Prior studies have shown the effectiveness of LEGO-based activities to improve children's social skills, especially those with autism spectrum disorder (ASD). In such activities, teachers are expected to provide instructions/guidance and facilitate communication among children during their collaborative play. When the ratio of children to teacher increases, an intelligent monitoring system could be very helpful to prompt the teachers of any necessary intervention, or to provide children some trivial feedback so that their teachers could focus on other tasks. In this paper, we propose such a system to assist both teachers and children in playing with LEGO bricks. At first, teachers may use the system to create some models. The system will then generate step-by-step instructions in both visual and auditory forms. In addition, the system can track the children's brick-building process in real time and provide some feedback when any mistakes were made. Two cameras with a unique image recognition module are used to capture the 3D structure of LEGO bricks.

CCS Concepts
• Human-centered computing → Accessibility systems and tools • Computing methodologies → Visual inspection • Applied computing → Computer-assisted instruction

Keywords
LEGO therapy; intelligent tutoring system; autism; children

1. INTRODUCTION
LEGO bricks are one of the most common toys among children. In this paper, we proposed a prototype of intelligent tutoring system to help young children with ASD (aged 3 to 5) to play with LEGO bricks. The system can recognize the structure of bricks using two cameras and monitor the children's work in real time. The teachers may use a separate terminal to create new structures (models) and the system can automatically generate the corresponding instructions in both visual and auditory form for each building step. This work extends our previous work reported in [1]. In the previous work we used a camera to recognize LEGO models from one point-of-view (POV) only. In this work, we utilize two cameras so that it can recognize 3D LEGO models.

At the current stage, the system can recognize and provide feedback when a user has completed all tasks or made a mistake in the meantime. In the future, more complex brick combinations will be supported, and more modalities will be designed for the users, e.g. speech recognition, motion, tactile, etc.

2. RELATED WORK
Children with ASD, as young as 18 months age, could benefit from early intervention [5, 6]. LEGO bricks, which are very popular among children, including toddlers, have been used as a therapeutic medium to enhanced children's social communication skills [2, 3, 4]. One of the reasons to use it is the predictability in constructing an assigned structure, which could make children with ASD feel more comfortable than playing in a free-play environment. To promote communication, teachers may assign children with various roles so that they will collaborate to build a structure following a set of clear instructions. Meanwhile, teachers may also intervene by helping and encouraging them.

Recently, technology-based intervention using LEGO play, including LEGO robots, have also been introduced to children with ASD [7, 8]. Using machine learning technology, some hobbyists have also developed LEGO brick sorting machines which could recognize and classify various bricks. However, using LEGO in an intelligent tutoring system for improving the social communication skills of children with ASD is rare, which motivates our study here. In the next section, we will describe the functionalities of our proposed system, its design, and architecture.

3. SYSTEM DESIGN AND CORE MODULES
Following our previous system in [1], our current system consists of two separate modules: a tutor's module and a child's module.

The tutor's module allows teachers/parents to design a new scheme/model of LEGO bricks, which consists of its construction steps. At this stage, our design aims for LEGO Duplo® bricks, and the maximum dimension of the model is 20×20×20 cm. Indeed, a larger model or smaller pieces could be supported when better cameras with higher resolution are used.

The child's module will load an assigned model, generate instructions for the children, and monitor the building process automatically. Ideally, both visual and voice instructions will be used to guide users in constructing the model.

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When a mistake was made by a user, the system shall detect it, remind the user (e.g. “not the green one”), and provide an instruction to correct it (e.g. “replace it with a blue one”). At current prototype, all instructions are provided by the application designer and cannot be customized.

3.1 Setup and Working Environment

As shown in Fig. 1, two cameras are put at the front- and right-side of the user, facing the working area. The visual instructions for the user are shown on the screen. The user needs to complete the assigned LEGO bricks in the working area. The computer processes the frames from two cameras, figures out the current 3D structure of the bricks, compared them with the goal, and generates instructions and feedback for the user.

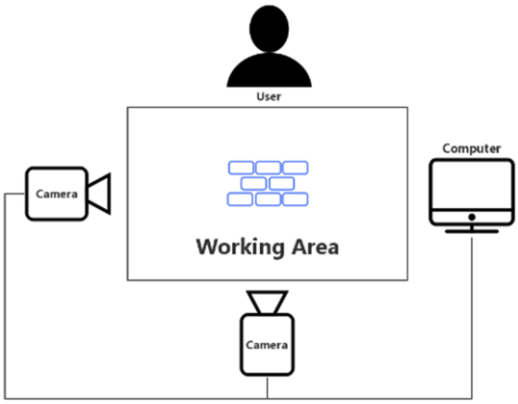


Figure 1. The setup of the tutoring system.

To achieve the best accuracy of brick recognition, the ambient light should be bright enough and the cameras should have a minimum resolution of 1280x720 pixels.

3.2 Brick Recognition Module

Currently, our proposed system can recognize three different shapes (long, short, curved), and three colors (red, green, blue) from basic LEGO Duplo® bricks, as shown in Fig. 2. To determine the shape of a brick, we use the ratio of its length to its height. For example, the long brick has ratio approximately 3.3:1, and the short one is 1.6:1.

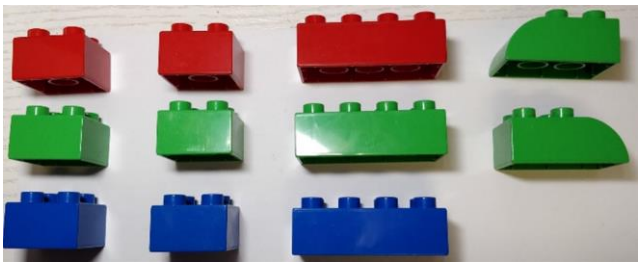


Figure 2. A set of bricks recognized by our system. [1]

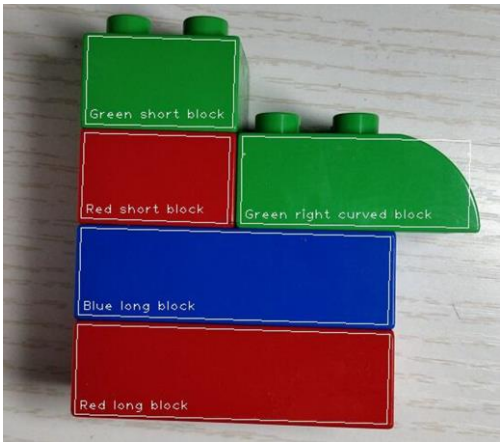


Figure 3. A set of annotations generated from one camera’s frame. [1]

At first, the system separately annotates the bricks from 2D frames captured by each camera, then synthesize two sets of annotations together to detect the shape of bricks. For example, if one camera annotates a brick with ‘long brick’, and the other camera annotates the same brick with ‘short brick’, then the real shape of the brick should be a long one since its planar projection shows differently on the two sides. Fig. 3 shows a set of annotations generated from one camera only (reported in [1]).

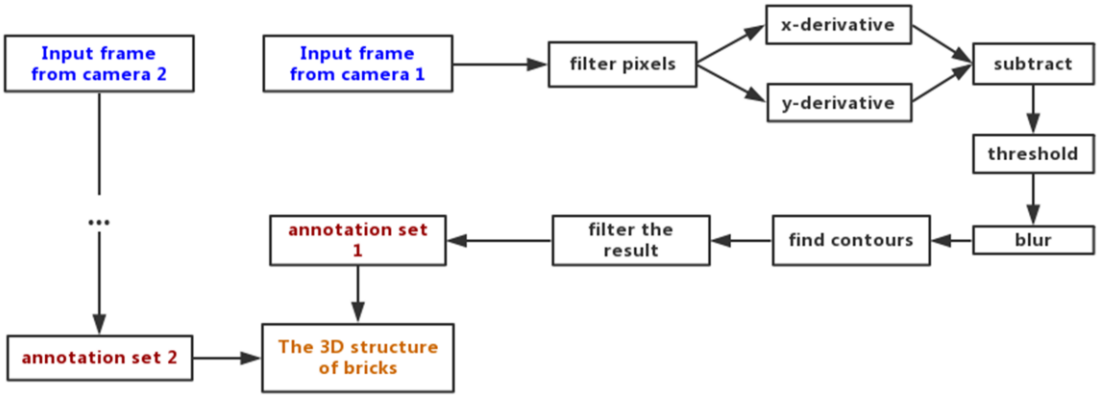


Figure 4. The process of frames from two cameras for deciding the 3D structures.

Fig. 4 shows the flowchart of how the 3D structures of bricks are recognized. To annotate the frames, our proposed system uses several functions in OpenCV library to process the frames and generate brick contours [9]. First, the pixels in a frame is filtered based on their RGB values, and the pixels with a color other than or not closed to red, green, and blue will be replaced with a white-color pixel. Then, the system computes the derivatives of each frame on both x- and y- axis using an extended *Sobel* operator and subtract them to make the contours clearer. Noises are then removed by using threshold-based filters and blur functions in OpenCV. A grayscale image will be sent to OpenCV's *findContour* function. For each contour from the result, a minimum rotated rectangle that contains the contour will be calculated. The system finally filters the rectangles based on their areas, sloops, length-width ratios, and positions. The remaining rectangles will be considered as bricks and annotated respectively.

3.3 Layout Recognition Module

To represent each brick's position in a 3D space, the program uses a unique coordinate system. The coordinate system annotates the 3D structure of bricks thus allows the comparison between different brick layouts. As shown in Fig. 5, the system fits the bricks into three axes. The x, y values depend on the left corner's projections on each axis, and the z value is based on the number of layers. Furthermore, for non-square bricks, symbol '-' or '|' are used to represent a brick which is parallel or vertical to the x-axis, respectively. For instance, the brick structure shown in Fig. 5 can be annotated in text by (assuming the green brick is a short one):

long blue 0 0 1 |

short green 2 0 1

long red 0 0 2 -

The annotations are generated after the brick-recognition process, and the result will be compared with the model assigned by the teacher to determine the correctness of the built.

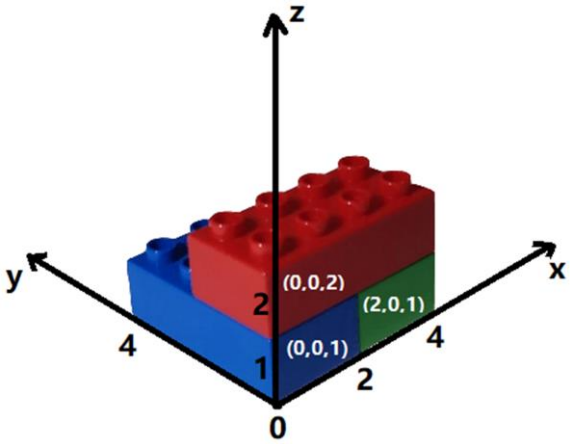


Figure 5. The coordinate system to represent bricks' positions.

4. CURRENT PROTOTYPES

4.1 The Tutor's Module

The tutor's module is designed for teachers to generate their LEGO models. Fig. 6 shows the user interface of the software, in which teachers can select and put the supported bricks to the grid on the left part. Then they can click 'output scheme' to save the

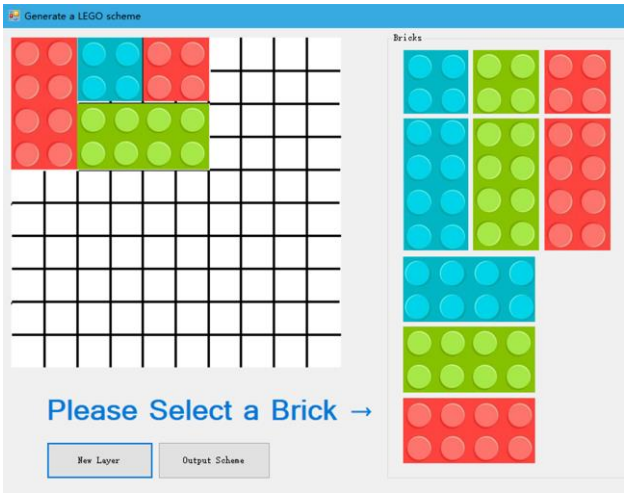


Figure 6. The user interface of the tutor's terminal.

structures as a completed model into the computer and, later, assign it as a task to the children. Since only two cameras are used, teachers are advised to put the bricks from the left-upper corner to the right and bottom, so that each additional brick could be recognized or seen by either or both cameras.

4.2 The Child's Module

Fig. 7 shows the user interface of the child's module. The interface is built using *cvui* library [10]. Two cameras track the bricks in real time and their frames are shown on the two left windows. When a teacher loads a new task, the system will automatically generate a preview window for the children and the subtasks. At each subtask, instruction on which brick should be picked will be given in both visual and auditory form. The current subtask is labelled with a star on the preview window (see the right window in Fig. 7). When the child correctly put a brick, voice feedback 'Great!' will be played and the corresponding brick in the preview window will be ticked. The voice feedback is generated using the text to speech service provided by Baidu™.

As shown in Fig. 7, the system asks for a red long brick, but the

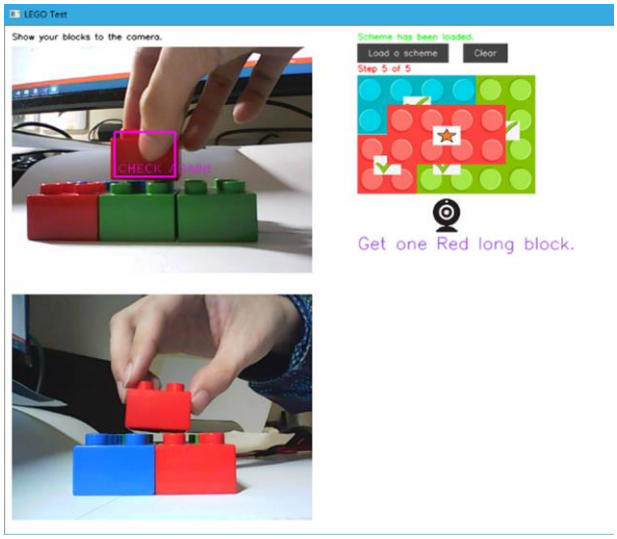


Figure 7. The user interface of the child's terminal.

user picks a red short brick. In this case, the mistake is automatically marked in the front camera's view, and a notification 'check again' appears.

5. DISCUSSIONS

In summary, we have proposed an intelligent system to help children with ASD in playing with LEGO bricks. The system is an extension of our previous work reported in [1]. The system can generate step-by-step guidance for a model defined by a teacher. Although our current system can accurately recognize a certain number of LEGO structure, some combinations may not be recognized well, even using two cameras. For instance, when a center brick is surrounded/covered by others, then it may be unseen from both camera POVs. To solve it, we may install additional cameras; however, they may increase the computational complexity of the system. Alternatively, we may restrict the teachers from designing a model which consists of such subtasks.

In the future, improvements will be made to increase the compatibility of different structures. Also, more interactive methods and feedback will be considered to make the system more interesting to be played with.

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